

Game of Chocolates

Difficulty: Medium | Accuracy: 45.81% | Submissions: 10K+ | Points: 4 | Average Time: 20m

Bunty and Dolly are playing a game described as follows:

- There are two boxes containing a and b chocolates respectively.
- In one move, a player can eat L ($L \geq 1$) chocolates from either one box or from both boxes simultaneously. Note that if a player picks from both boxes, then the number of chocolates picked should be same.
- The players play alternately, and the player who eats the last chocolate wins.
- Both Players want Dolly to win.

Find who should play first to make Dolly win. Return true if Dolly should play first; otherwise, return false.

Examples

Input: a = 1, b = 2

Output: false

Explanation: If Bunty starts first, all the possible states after his move are (0, 2), (1, 1), and (0, 1). These are all winning states for Dolly. Therefore, Bunty should play first.

Input: a = 1 and b = 3

Output: true

Explanation: If Dolly starts first, all the possible states after Dolly eats chocolates are (0, 3), (1, 2), (1, 1), (1, 0), (0, 2). Out of these, (1, 2) is the winning state for Dolly.

Constraints

$0 \leq a, b \leq 10^9$

Expected Complexities

Time Complexity: $O(n)$

Auxiliary Space: $O(1)$

Topic Tags

Game Theory

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